A Novel Unified Trilateration Method for RSSI based Indoor Localization

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***Abstract*— Indoor positioning exploiting RSSI has been widely used due to the advantages of its ease of installation and low cost among lots of indoor localization technologies. Trilateration, which is commonly employed to estimate the location in such an indoor environment, has a constraint that three circles must meet at one point, so many modification algorithms have been studied to compensate for this. However, in most of the existing methods, different algorithms are individually applied to estimate the location for each of the various cases where the three circles do not meet at one point. In this paper, unlike conventional methods, we propose a new unified trilateration method that can be consistently applied to all cases whether the three circles meet at one point or not. We verify the performance of the proposed algorithm through the Monte Carlo simulation.**

Keywords—RSSI, Trilateration, Indoor Localization

I. Introduction

As untact platforms are receiving a lot of spotlight due to the recent COVID-19, indoor localization technology is necessary to provide increasing services such as IoT, VR/AR, AI, etc. Relatively accurate positioning using GPS (Global Positioning System) is possible in outdoors, but such signals become unstable in indoor environments due to the roof and walls of buildings, resulting in a large error in localization. Techniques used for indoor localization include RSSI (Received Signal Strength Indicator), ToA (Time of Arrival), TDoA (Time Difference of Arrival), CSI (Channel State Information), PoA (Phase of Arrival), AoA (Angle of Arrival), etc [1]. Among them, indoor localization exploiting RSSI, which has low complexity and easy to implement, has been widely studied and utilized [2][3].

For RSSI based indoor localization, the RSSI is converted into distance value, and the *Log-Distance Path Loss Model* is typically used due to its simplicity [4][5]. However, in the real environment, when converting to a distance value, noise is mixed due to the influence of multi-path fading and ambient interference signals, so that it is not feasible to obtain an accurate distance value [6],[7]. Due to this, as shown in Fig. 1, three circles do not meet at one point in trilateration, so many techniques have been considered to compensate for such cases [4]-[9].

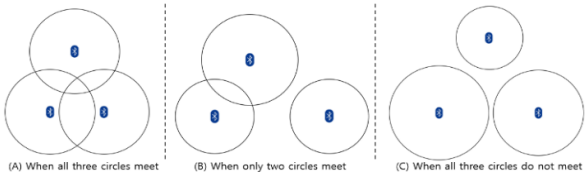


Fig. 1. Example when three circles do not meet at a point in trilateration

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In the conventional approaches for trilateration, when the three circles do not meet at a point, different correction techniques have been applied to each case. In this paper, unlike the existing methods, we propose a novel location estimation method that can be consistently applied to all cases whether the three circles meet at one point or not. We will verify the performance of the proposed algorithm through the Monte Carlo simulation

II. A Novel Unifided Trilateration

A. Log-Distance Path Loss Model

For indoor localization using RSSI, it is required that a RSSI value is converted into a distance value. In general, the Log-Distance Path Loss Model[4][5] is widely used, and also adopted in this work, which is be expressed by

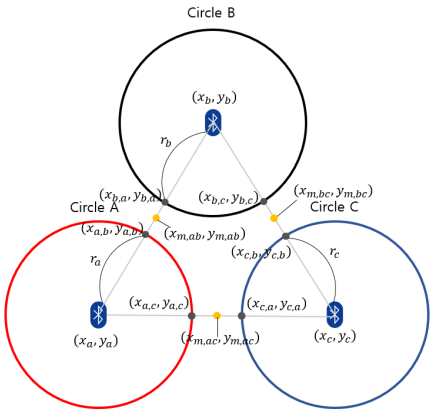
$$\mathrm{RSSI} = \mathrm{A} - 10 \cdot \mathrm{n} \cdot \log\left(\frac{\mathrm{d}}{\mathrm{d}_0}\right) + \mathrm{X}_\sigma \qquad (1)$$

where RSSI is the received signal strength at distance d, and its unit is dBm. A is the received signal strength at d_0, which is the reference distance (1m is generally used for d_0 .) And n means the path loss exponent, and it is determined through actual measurement because it differs depending on the environment. X_σ is noise mixed into the signal, which also varies depending on the environment, and it is usually represented as a *Gaussian* random variable. So there are also algorithms for calculating these constants according to the environment [10][11]. However, in this paper, fixed values obtained through actual measurements are used. The formula to convert RSSI to distance value using (1) is as follows.

$$\mathrm{d} = \mathrm{d}_0 \cdot 10^{\mathrm{A-RSSI+X}_\sigma} \qquad (2)$$

The location is estimated by applying a trilateration algorithm to the distance value obtained using (2).

B. Proposed Unified Trilateration Method



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Fig. 2. The points where each circle intersect with the line connecting the centers of each circle, and the midpoint between the two points

A novel unified trilateration proposed in this paper constructs a triangle using points on each circle and uses the incenters of the triangles. It can be applied to the case where all errors occur, but for ease of explanation, the case where all three circles do not meet (Fig. 1 to (C)) will be described as an example.

Circle A and circle B are respectively expressed by

$$(x - x_a)^2 + (y - y_a)^2 = r_a^2 \quad (3)$$

$$(x - x_b)^2 + (y - y_b)^2 = r_b^2 \quad (4)$$

And a straight line passing through the respective centers of circle A and circle B can be obtained as follows:

$$y = \frac{y_b - y_a}{x_b - x_a} x - \frac{y_b - y_a}{x_b - x_a} x_a + y_a \quad (5)$$

In order to find the point where this straight line and circle A and circle B meet, respectively, (5) is combined with (3) to find the point $(x_{a,b}, y_{a,b})$ where circle A and the straight line meet. And (5) is combined with (4) to find the point $(x_{b,a}, y_{b,a})$ where circle B and the straight line meet. And find $(x_{m,ab}, y_{m,ab})$, the midpoint between those two points. In the same way, when circles B and C are the standards, when circles A and C are the standards, $(x_{m,bc}, y_{m,bc})$ and $(x_{m,ac}, y_{m,ac})$ are respectively calculated.

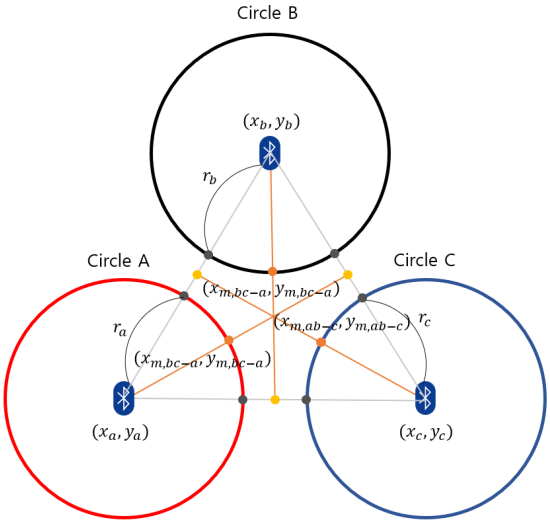


Fig. 3. A straight line passing through each midpoint and the center of the opposite circle and the intersection point of the straight line and the circle

Next, the equation of a straight line passing through $(x_{m,ab}, y_{m,ab})$ and (x_c, y_c), the center of circle C, is given as :

$$y = \frac{y_c - y_{m,ab}}{x_c - x_{m,ab}} x - \frac{y_c - y_{m,ab}}{x_c - x_{m,ab}} x_{m,ab} + y_{m,ab} \quad (6)$$

As we can see Fig. 3, circle C and (6) is combined to find the intersection $(x_{m,ab\text{-}c}, y_{m,ab\text{-}c})$. In the same way, we get both $(x_{m,bc\text{-}a}, y_{m,bc\text{-}a})$, $(x_{m,ac\text{-}b}, y_{m,ac\text{-}b})$.

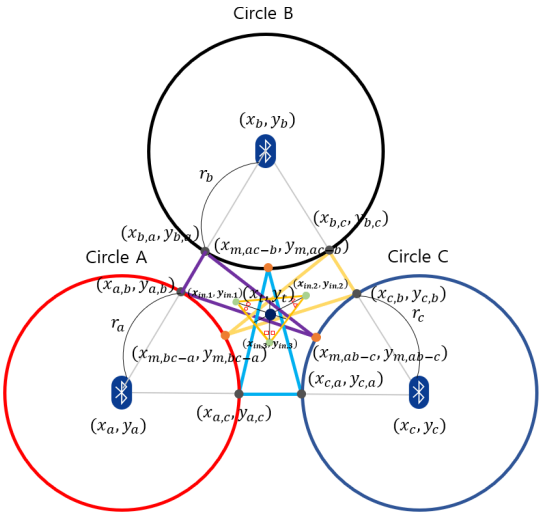


Fig. 4. The incenter of each triangle and the incenter of a triangle connecting the incenters

As a last step, in Fig. 4, draw a triangle with $(x_{a,b}, y_{a,b})$, $(x_{b,a}, y_{b,a})$, and $(x_{m,ab\text{-}c}, y_{m,ab\text{-}c})$ as vertices, and find the incenter of the triangle $(x_{in,1}, y_{in,1})$. In the same way, $(x_{in,2}, y_{in,2})$ and $(x_{in,3}, y_{in,3})$ are obtained, and finally, (x_t, y_t), which is the incenter of a triangle having these three incenters as vertices, is obtained and 소 is coordinate value is determined as the final position of the target to be estimated.

III. Simulation Results and Discussions

A. Simulation Environments

In RSSI, since the n value, which is the path loss exponent, and σ, the standard deviation of the Gaussian random variable X_σ in (1), vary greatly depending on the actual measurement environment, it is judged that an experiment through simulation can obtain an objective result, so simulations are performed to evaluate the accuracy. We compare the performance of the algorithm proposed in this paper with Bias-Reduced Maximum Likelihood (BML)[4] and Weighted Trilateration[8]. Our simulation is conducted based on MATLAB, and *fminsearch* tool is used to implement BML.

The size of the indoor environment is set to 10m×10m and 20m×10m, and 4 beacons are used. The reason why it is set to 10m×10m and 20m×10m is because it could be observed that reliable RSSI could be obtained mainly within 20m distance through the actual beacon RSSI measurement experiment. In addition, it is assumed that the interference and noise caused by the surrounding environment experienced by the signal of each beacon are applied independently. Here, interference and noise are assumed to be log-normal distributions with mean 0 and variance σ^2 , and the performance of the proposed algorithm is compared and verified according to this variance value, σ^2. In the simulation, the n value, which is the path loss exponent, is set to 3.5, and the A value is set to -55, and it is assumed that the location of the target to be positioned is randomly located in the indoor space.

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B. Scenario I

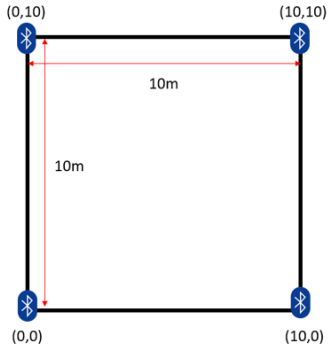


Fig. 5. Simulation Environment for Scenario I

In simulation scenario 1, as in Fig. 5, an indoor environment of 10 m × 10 m is considered. Four beacons are placed at the coordinates (0,0), (0,10), (10,10), and (10,0), respectively, and simulation is performed.

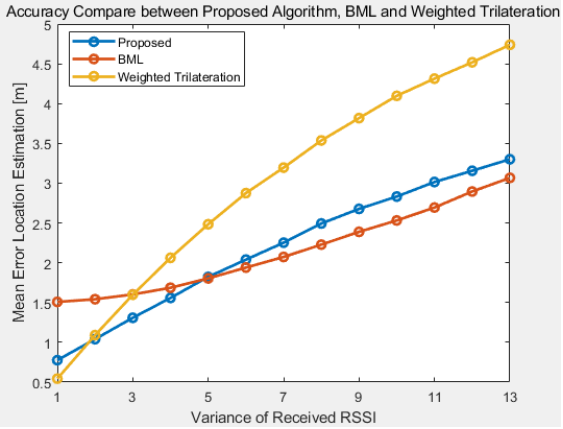


Fig. 6. Mean Estimation Error for Scenario I

Figure 6, when the variance is very low, the performance of weighted trilateration is better, but in an actual indoor environment, the variance of RSSI is rarely very low. In most cases, we find that the algorithm proposed in this paper outperforms the Weighted Trilateration often referenced in trilateral measurements used for indoor localization. It is observed that there is not much difference in performance from BML.

C. Scenario II

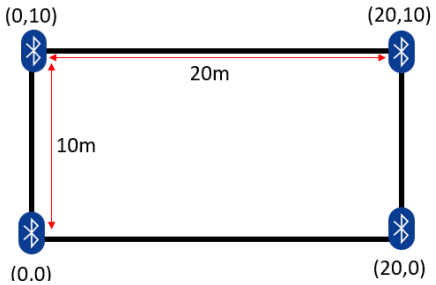


Fig. 7. Simulation Environment for Scenario II

In simulation scenario II, as in Fig. 7, an indoor environment of 20 m × 10 m is considered. Four beacons are placed at (0,0), (0,10), (20,10), and (20,0) positions, respectively, and simulation is performed.

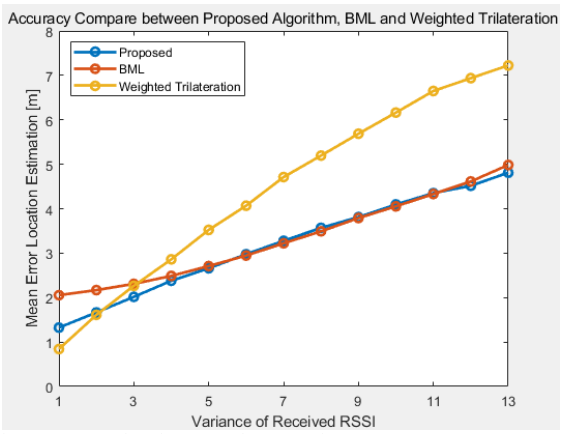


Fig. 8. Mean Estimation Error for Scenario II

From Fig. 8, same as in scenario 1, it can be found that the algorithm proposed in this paper has higher accuracy than Weighted Trilateration in simulation scenario II. When compared to BML, the proposed algorithm has higher accuracy when the variance is 5 or less, but when the variance is 5 or more, it could be confirmed that they are similar.

IV. CONCLUSIONS

In this paper, we proposed a novel unified trilateration algorithm that can be applied consistently in all cases, unlike the existing trilateration method in the RSSI-based indoor localization system. By using the proposed technique, it is possible to not only increase the positioning accuracy, but also to avoid applying different algorithms to each error cases like the existing trilateration method. This greatly reduces the implementation complexity. Through simulation, it is verified that the accuracy is much higher than that of the Weighted Trilateration method, which is widely used among the existing trilateration methods, and it is also confirmed that the performance is similar to BML, one of the ML-based algorithms with the highest accuracy in the indoor localization method.

ACKNOWLEDGMENT

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